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Financial Services

Architecting the IT Organization: On IT Architecture Tools and Techniques and Their Application to (IT) Organizational Design

EXECUTIVE SUMMARY

How should the IT organization be designed and operated? BearingPoint explores 10 approaches to addressing IT governance issues. Our hypothesis is that we can leverage what architects have learned about analyzing and resolving problems that arise in dealing with any kind of complex system.

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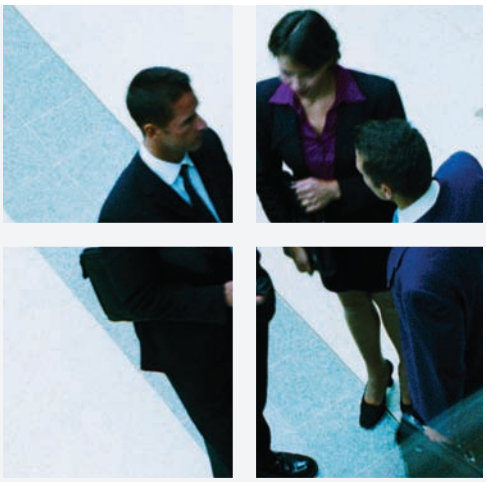


TYPICAL IT ORGANIZATIONAL DESIGN ISSUES

Typically, IT organizational structures are the result of an evolutionary process and are not necessarily the outcome of planned design. In many cases, IT organizations came into being as the outgrowth of either the financial or facilities functions with entirely different requisite skill sets. For the past 10 to 15 years, a totally new paradigm has been needed because the typical IT organization is still hindered by a number of legacy decisions leading to sub-optimal design:

Typically, IT organizational structures are the result of an evolutionary process and are not necessarily the outcome of planned design.

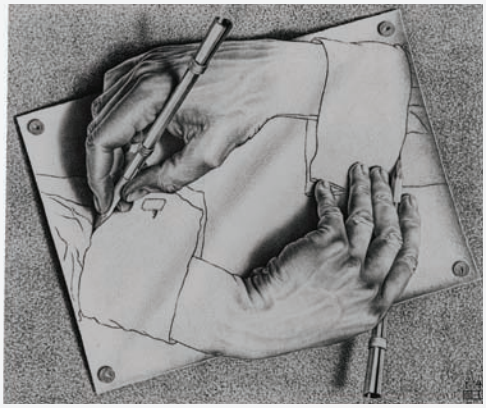
- A number of organizations, particularly in the financial services industry, are the product of many mergers. After a merger, and this is true for IT and every other function, certain principles are invoked, ranging from the "let's leverage leading practices from both camps" approach to "let's repaint everything in the colors of the acquiring company's logo."

  But often neither of these "leading-practice/merger-of-equals" or "takeover" scenarios really happens. Instead, whole portions of the resulting merged organization are assigned to executives of the legacy organizations through a protracted process that protects some individual careers without too much logic, aside from the desire to define a "politically balanced" organization.
- When various IT groups are consolidated, there is very frequently no conscious attempt to deal with the necessary responsibility overlaps and redundancies beyond a superficial aggregation of activities. This issue is particularly true at the executive level and in situations involving outsourcing when organizational charts continue to include functions and people that presumably need to be outsourced.
- Complex organizational governance models are built around personalities instead of the result of a deliberate process. Typically, technologists, but also non-technologists, are put in charge of managerial duties for which they are not necessarily prepared.

Finally, to compound the problem, solutions for these issues are often implemented without using or invoking explicit principles or actual facts. This is particularly problematic when dealing with an audience of technologists who tend to respect and promote systematic process and adherence to standards and methodologies.

FIGURE 1 *Drawing Hands*, M.C. Escher, 1948: Designing the IT Architecture Function Can Be Done Using Architecture Tools and Constructs

Questions such as where, how and by whom should architecture be defined can be elegantly answered using the very principles employed to define a technology architecture itself—a duality that is well illustrated in Figure 1, *Drawing Hands* by M.C. Escher.



As a result, when attempts are made to mold the organization around existing people and skill sets, employees impacted by the change not only begin to question the decisions made but to resist their implementation while demanding to know how decisions were made and how fair the process has been. This invariably leads to situations in which management goes through a "decide and announce" or "decide and sell" approach, with neither succeeding in rallying employees to support change.

There must be a better way.

THINKING LIKE AN ARCHITECT

Designing an IT organization, or any organization for that matter, should be the product of a systematic analysis that defines the bridge between the business vision, organizational constructs, processes and enabling technologies required to execute the vision.

The organizational architecture is the foundation upon which decisions are made, processes are developed, and supporting technologies are selected and implemented so that business processes are served.

So here is the challenge: How do you resolve the system of simultaneous equations that define each component of the model and determine how they interact together? Moreover, how do you accomplish all of this with the typical "boundary conditions" all organizations face—for example, the lack of complete or even partial understanding of either the current state or uncertainty as to what the end state should be. Architects have to deal with this level of complexity and uncertainty all the time.

In the sidebar, "A Universal Architecture Toolset," are several of the approaches regularly leveraged in architecting technology solutions. These approaches can be used in architecting organizations, too. Each of these approaches is essentially transposable "as is" from the IT architectural realm to organization architecture.

We will review each of these approaches in turn with a brief description of the underlying concepts, examples of use in the technology realm, and a description of how to use this approach in the context of organizational design and transformation.

The concepts (some would say the "meta-patterns") are presented in a logical order so that one concept can be explained by previously introduced references but the order does not reflect the relative degree of importance of each tool, which should be viewed as a component of an overall toolset.

1. UNDERSTANDING "CEREMONY" AND THE "ITERATIVE/WATERFALL" TRADE-OFFS

What Is It?

Ceremony. Any organization attempting to undertake a technology-oriented project must choose—or must come to realize in the course of the project iteration—what level of formality will be used in the course of the project. Here are some examples:

- How detailed should the documentation be? The question pertains not only to the deliverables associated with the project but also to other documentation that describes the project mission, the assumptions made and other aspects of the project.
- How many formal status meetings should there be per day/week/month? What should the attendance be and how formal should the documentation be?
- With regard to communication to the executive sponsorship, how often should they be brought back into the loop? Should communication be via periodic e-mails, using formal deck presentations, or informally as the project goes?

Altogether, this degree of formality and detail in communicating and documenting activities with team members and with the various project stakeholders—for example, management, business units impacted and vendors—is encapsulated in the concept of ceremony.

"High ceremony" refers to an approach with emphasis on formality in communication, a high degree of details in reporting progress and low frequency of change of documentation. "Low ceremony" refers to an approach that focuses on informal, more frequent communication using lightweight documentation that changes frequently.

Waterfall versus iterative. A number of situations lend themselves to a linear solution approach, by which we mean that a project team typically starts by defining the issues, identifying possible approaches, selecting the optimum one based on predefined criteria, and then implementing and testing the results for consistency with initial assumptions. This is known as a waterfall approach—the symbolism points to the linear and somewhat inexorable flow of cascading activities, as well as the existence of defined transition points in the approach.

A UNIVERSAL ARCHITECTURE TOOLSET

Architectural practices and concepts that can be transposed to organizational design:

1. Understanding "ceremony" and the "iterative/waterfall" trade-offs
2. Using "views": Looking at issues from several angles
3. Understanding the difference between the conceptual, logical and physical views of the issues at stake
4. Using value chain analysis
5. Putting the value chain in motion: Leveraging use case analysis
6. Distinguishing between principles, standards and implementation considerations
7. Using patterns to document and leverage what works and avoid what does not
8. Using agile documentation and lightweight planning: The importance of plateau definition
9. Embracing service orientation[1]
10. Leveraging other toolsets as you see fit

[1] This item is a play on words that attempts to draw an (admittedly tenuous) relationship between the service-oriented architecture (SOA) in IT architecture and another concept in the organizational architecture that promotes the definition and use of product management concepts in the definition and provisioning of IT-based solutions.

By reflecting one style, project teams implicitly or explicitly decide to use a methodology along the "iterative or waterfall" and "high or low ceremony" continuum.

This contrasts with a perhaps less intuitive, but at times more powerful, approach in terms of how fast one can get to results and how good the results are. In the so-called iterative process, there is a constant attempt to retrace your steps and improve on previous results through incremental changes. This iterative approach can be taken at various levels—for example:

- **Project level iteration.** A project team might decide to develop a first release of a solution with limited features, knowing full well that this will not be the final result. Each successive iteration of the activities is then built on the results of the previous one until the project is declared complete because the current result is good enough to be accepted by the stakeholders.
- **Task level iteration.** The same approach can apply at the task level. For example, a document evolves and each draft incorporates new information.

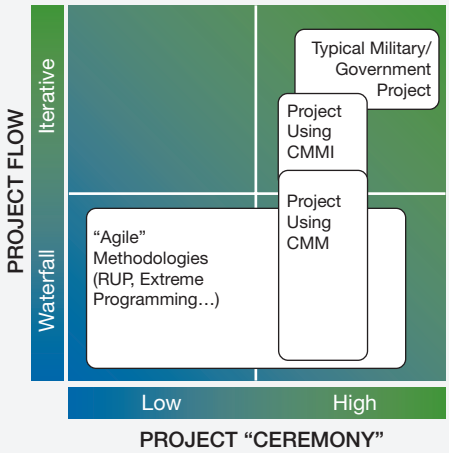
Examples in the IT World

By reflecting one style,[2] project teams implicitly or explicitly decide to use a methodology along the "iterative or waterfall" and "high or low ceremony" continuum, as described in Figure 2.[3]

A more complete discussion of the relative merits and degree of use of the various combinations of ceremony and flow is out of the scope of this paper, but a few remarks are in order:

- High ceremony using the waterfall approach has been a common, essentially de facto, approach in large projects, particularly those mandated by governmental agencies, which have actually published standards, such as DOD-STD-2167A, MIL-STD-498, where a high level of risk management and traceability of results are demanded.
- This has also been traditionally used by systems integration departments with similar objectives or enterprise resource planning (ERP) system implementation teams, particularly if the projects are distributed globally and impact various business units. This is where adherence to and compliance with a well-defined assessment framework, such as Software Engineering Institute's CMM,[4] is favored—for example, to make sure that teams developing a piece of software using a combination of on- and offshore resources communicate and work well together.

FIGURE 2 Strategy Development Process



[2] The term *style* is used here to describe the combination of ceremony and use of iterative process.

[3] Source: Per Kroll and Philippe Krutchen, *The Rational Unified Process Made Easy*, Addison Wesley, 2003, ISBN 0-321-16609-4, page 53.

[4] Capability Maturity Model.

- A variation of CMM, CMMI[5] incorporates some of the leading practices used in the iterative processes, particularly by suggesting that team members focus selectively on areas of highest return, in terms of benefit to end users and risk management.
- Agile methodologies have emerged in the past decade. They use low ceremony and the iterative processes for software development. The main characteristic of these approaches is that they provide a great level of flexibility that is particularly well adapted to situations in which requirements are either not very well defined or are evolving, or when the environment itself is evolving rapidly.

  Considering the constant acceleration of business cycles, these types of approaches have been very adaptable in the current business environment where long term essentially means "within the current budgetary cycle."

  But there is more to agile development processes. Their success stems from their essential pragmatism and the fact that value is delivered very early and increases at each iteration. High ceremony and waterfall approaches assume and demand a perhaps unrealistic type of stability in the user environment—for example, requirements not changing for the entire duration of what are sometimes years of development—or users' patience in getting something in their hands at the end of a long, mostly opaque process. At that point, trying to make change is too late.

High ceremony and waterfall approaches assume and demand a perhaps unrealistic type of stability in the user environment—for example, requirements not changing for the entire duration of what are sometimes years of development—or users' patience in getting something in their hands at the end of a long, mostly opaque process.

Application to IT Organizational Design

Envision your first day in leading an IT transformational change focusing on organization. Whether it is a wholesale multiyear project or the redesign of the inner workings of a single department, you will have to decide what will work best.

For large, enterprisewide projects, there is probably no other way but to adopt a high level of formality and linearity in execution. If employees are widely dispersed geographically, communication has to be done with some level of ceremony, not only because of practical considerations—a lot of one-to-many communications are needed— but also because the potential for ambiguity increases exponentially with the size of the project. This requires a substantial amount of well-calibrated, scripted communication.[6]

For example, reorganizing the infrastructure component of an IT group—that is, the one dealing with network and data centers—might take several years just because of the sheer size of the project, so there is no practical alternative to executing change using a waterfall approach.

[5] Capability Maturity Model Integration.

[6] This is where the Six Sigma DMAIIC (Define Measure Analyze Improve Implement Control) problem-solving approach works well. See the description of this concept in section 10 of this point of view.

Overall, the agile-iterative approach (agile) is more suited to organizations with highly competent and self-motivated individuals.

Similarly, a group with a strong engineering culture might be more inclined, and in fact might demand, an approach that states objectives upfront and shows a road map toward that goal with a great level of granularity—high ceremony and waterfall.

But this is not a general rule. More and more engineering-oriented organizations have embraced agile development methodologies. Those organizations might be inclined to consider a similar approach to organizational transformation based on incremental improvement with a lot of one-on-one consultation, as opposed to the more formal use of "all hands" meetings — that is, the use of less ceremony. Look for this type of change if similar approaches using other means—for example, (consultant-driven), high ceremony, waterfall approaches—have failed before.

Finally, large and complex does not necessarily imply high ceremony and waterfall (Case Study 1).

Overall, the agile-iterative approach (agile) is more suited to organizations with highly competent and self-motivated individuals. These individuals also tend to be averse to "strict" methodologies and the associated rigors, and they are better suited to the more "laissez-faire" attitude encouraged by agile. To that end, organizational designs should leverage a rapidly changing agile-based approach primarily in the setup of Engineering Centers of Excellence (COE)—further discussed in section 4, "Using Value Chain Analysis," where highly competent individuals address complex architectural issues, evaluate cutting-edge technologies and their relevance, and develop reusable components and frameworks. The work done here is then leveraged across the enterprise.

From a communication standpoint, various communication styles might coexist:

- Typically executives need (or rather want without necessarily questioning why they do so) to be presented with information with a high degree of formality/ceremony, including a formal presentation as to what the return on investment[7] is projected to be, along with a high-level project plan showing critical milestones.
- Employees impacted by change need to be provided with more than an abstract project plan. They need and deserve to understand how a solution will be arrived at through collaborative iterations. In fact, they need to be convinced that the whole process will be iterative in nature and that the results of each iteration will be presented to executive management as a milestone.

[7] IT organization architecting projects, just like IT architecting projects, are hard to justify on the exclusive basis of return on investment analysis, except when the project is essentially geared to cutting costs through job elimination.

Large and complex does not necessarily imply high ceremony and waterfall.

CASE STUDY 1: USING AN AGILE APPROACH (LOW CEREMONY/ITERATIVE) TO IT ORGANIZATIONAL TRANSFORMATION

Here is a case study that shows how an enterprise-level transformation process was dealt with successfully, using an iterative, low ceremony approach to accelerate the use of offshore resources for application development.

A company with a substantial amount of activity focusing on the development of ERP and customized software solutions for third parties wanted to boost the use of offshore resources. In the past, the use of these resources had remained at a relatively low level since the firm established a few new development centers abroad.

The previous attempts to encourage the use of these centers have used intensive top-down encouragements, including highly visible and repeated announcements of the opening of the centers and even special internal pricing to encourage development groups to leverage the new centers. Executive management designed detailed, but ultimately unrealistic, growth plans dutifully agreed to by all but not implemented in practice.

The use of these centers only took off after one of the business unit leaders took a very different tack. He asked for a very small budget to create an internal technology interface group (TIG) to liaise with the centers and to subsidize some projects so that they could use that new TIG entity. He did not present a detailed ROI analysis. Nor did he suggest additional enterprise communication, recommending instead that a low-key approach be used. In particular, he insisted on personally marketing the use of the new TIG within the company. He also predicted correctly that the success of the project would be achieved through a form of viral marketing—i.e., that the success of the initial projects would attract other groups to use the TIG. In his own words, "We'll make this a skunk works group that will work through word of mouth."

What he called "creating the buzz" did indeed create iterative levels of success and significant growth in the use of offshore centers where a more formal approach had previously failed.

Every problem, whether it is related to technology or organizational design, can and should be considered along a number of relevant dimensions.

Summary of Key Ideas

- Understand the high and low ceremony and iterative/waterfall concepts and determine early on what will work best for your initiative.
- Do not assume that technologists only favor one approach over the other. Instead, assume they understand these concepts because they have to deal with them in the context of the execution of technology projects.
- Define and use this terminology with team members and stakeholders to understand their preferences but also to explain your approach and set expectations.
- Understand the various technology methodologies involved in each approach and their relative advantages. These methodologies involve technology-specific ones—for example, eXtreme Development—and others that are more general in nature—for example, Design for Six Sigma.[8]
- Be practical. At times, using different communication styles with different constituents for the same project might be more appropriate and effective.

2. USING "VIEWS": LOOKING AT ISSUES FROM SEVERAL ANGLES

What Is It?

Every problem, whether it is related to technology or organizational design, can and should be considered along a number of relevant dimensions. Architects, whether they are developing technology architectures or organizational architectures, constantly look at the issues from various angles to understand them fully. Each of these angles, or views, provides an insight into the overall problem space. Using views greatly simplifies the architecture definition process.

Examples in the IT World

Consider a project dealing with the deployment of an ERP package. To be successful, the ERP implementation needs to consider a multitude of views that describe the current state, the end state and the road map to that end state.

[8] See the description of Design for Six Sigma in section 10.

For example, you need to develop a description of the situation before, during and after the ERP implementation in the following domains: the user interface, the application logic, the data layer, and the underlying computing and communication infrastructure, as well as the interfaces with current systems and resources.

Each of these dimensions is critical and interdependent. But each needs to be looked at separately as well, using separate views for each domain. Each of these views must be consistent with one another and each is relevant to specific types of technologists with specific focus, such as end-user presentation architects and data architects.

Each of these dimensions is critical and interdependent. But each needs to be looked at separately as well, using separate views for each domain.

There are also other types of domains that are equally important and require separate views, such as governance, process and organization. In addition, as we will see in the next section, you can look at these domains at various levels of specificity, such as conceptual, logical and physical.

Combining these domains provides a convenient way of describing the scope of a particular set of issues and solutions of the overall problem space. For example:

- Application architects who develop the high-level structure of the ERP solution need to start looking at what the "conceptual x application architecture" view of the project is.
- IT operations people who are concerned about how the data centers will verify that the ERP solution is properly deployed need to consider issues pertaining to the combined "physical layer x process x computing infrastructure" views of the ERP project.

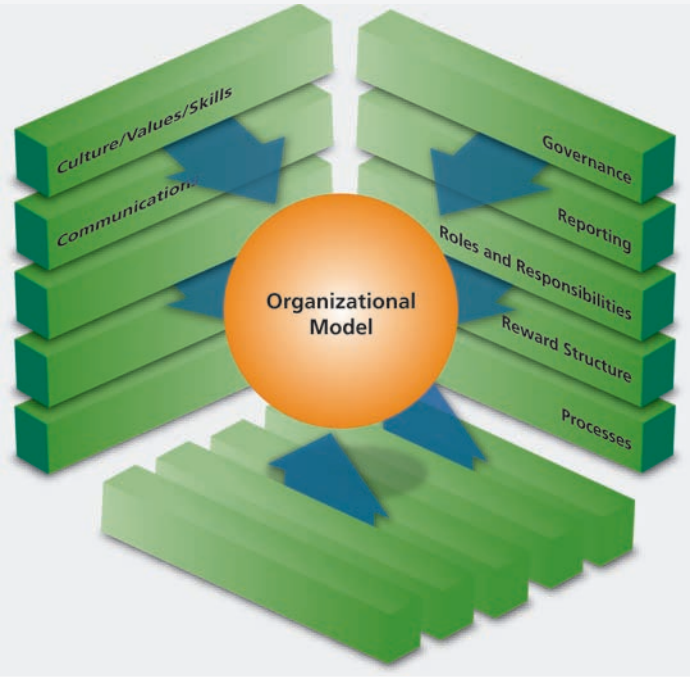
The overall architecture can be expressed as a series of views that help various audiences and stakeholders understand what the architecture is. It's important to note that each of these views is necessary, but once they have been recognized as relevant, each of them should be considered. One of the jobs of the chief architect is to identify what domains/views are relevant and to make sure that they are then taken into consideration and analyzed.[9]

Application to IT Organizational Design

Fundamentally, organizational constructs, processes and tasks, supporting tools, metrics, standards, and principles can be considered views of the overall model that need to be implemented with the appropriate tools on the right infrastructure and consistent with the organization's overall culture and value system. Other relevant views of the model also include governance, reporting, and roles and responsibilities (Figure 3).

[9] For the mathematically inclined, this is about projecting the n-dimensional problem space into subspaces of lower dimensions. Identifying the appropriate dimension of the space is about finding a free basis of the subspaces, ideally orthogonal.

FIGURE 3 The Concept of Views Applied to Organizational Design



Summary of Key Ideas

- IT solutions and IT organizational models are too complex to be considered as a whole. Approaching their design through various angles or views helps manage complexity. Views are interrelated but self-contained.
- For each problem, there are a number of views that can be considered. But whether designing an IT or organizational architecture, make sure you identify the views that are relevant to the environment.
- You can try to prioritize views, but be ready to review these choices later. For example, what might initially be thought of as a metrics problem might in fact be created by issues in values.

A useful way of organizing views is to consider issues at three broad levels of complexity and specificity: conceptual, logical and physical layers, as discussed in the next section.

3. UNDERSTANDING THE DIFFERENCE BETWEEN THE CONCEPTUAL, LOGICAL AND PHYSICAL VIEWS OF THE ISSUES AT STAKE

What Is It?

Architects segregate issues by making a distinction between the conceptual, logical and physical views of the issues at stake, from now on referred to as the "CLP" trilogy. Grouping architecture activities and work products at different levels of specificity allows architects to provide the right level of information to different users within the organization.

FIGURE 4 The Conceptual, Logical and Physical Levels of Specificity in Architecture and Who Uses Each of Them

CHANGE FREQUENCY			
Low	Conceptual Architecture	This specificity provides the highest level of abstraction. It defines the overall vision supporting the architecture and describes the fundamental principles that will be used for future IT decisions. It must be closely aligned with the business vision.	Management Layer Business Planners Executive Management Technology Strategists
Medium	Logical Architecture	This specificity is also called the "target architecture." It includes actionable models and engineering blueprints that differentiate approaches and provide a set of product-independent architectures for IT components not fully defined. It also includes a set of standards for technology domains.	Planning Layer IT Architects Product Managers
High	Physical Architecture	This specificity provides the highest level of detail and is also called the "implementable architecture." It includes physical engineering diagrams, detailed designs, "buy-in lists" and recommendations regarding the implementation of technology solutions and associated processes and tools.	Implementation/ Operations Layer Contractors Engineers Procurement Managers Implementers

The distinction between the various CLP views is immensely useful when developing technology solutions because it greatly simplifies the interaction between the various stakeholders involved.

Examples in the IT World

The distinction between the various CLP views is immensely useful when developing technology solutions because it greatly simplifies the interaction between the various stakeholders involved. The distinction confirms that shareholders get the information they need and are not encumbered by information they do not need.

For example (Figure 4):

- Corporate planners rarely need anything more than access to a conceptual view of the solution they are presented with for approval.
- Individuals concerned with the dissemination of knowledge within the IT organization so that effective and appropriate design practices (or patterns—see section 7) propagate within the organization need to develop and share logical architectures or blueprints.
- Finally, people involved in the actual implementation of the solution need specific plans and information, such as what is the recommended buy list, for their job without the high overhead that a more holistic description of the requirements would provide.

The distinction between the CLP layers also encourages that their contents are revised with the appropriate level of frequency. Conceptual architecture is usually established at the onset of a project for a specific initiative or only once every two or three years at the enterprise level, while physical architecture might change quite frequently to accommodate changes in technology used by the actual suppliers of hardware or software implemented.

Application to IT Organizational Design

Keeping the level of abstraction at the right level is key and there are many instances when managers need to use the CLP distinction.

Here again, keeping the level of abstraction at the right level is key and there are many instances when managers need to use the CLP distinction. For example, before rethinking an organization, managers should first agree on some overall description as to what the organization does, conceptually and logically.

Conceptual view. The conceptual view of the IT organization should succinctly encapsulate the general principles[10] presiding over the definition of the organization's mission, as well as the vision of the enterprise, as to what the IT organization should aspire to be and do. For example:

- Different organizations aspire to various degrees of self-sufficiency in controlling the resources and the IT value chain—that is, the set of processes associated with the production of solutions to the end users.[11] Options range from total insourcing to total outsourcing. Organizations also allow various degrees of discretion in having the business units make their own decisions to use or not use IT shared services.
- Organizations also exhibit various degrees of willingness to partner with vendors, as opposed to considering them exclusively as providers of parts, be it hardware, software or individuals used in staff augmentation mode.

The various combinations of options leads to conceptually very different types of organizations (see Figure 5 for two extremes):

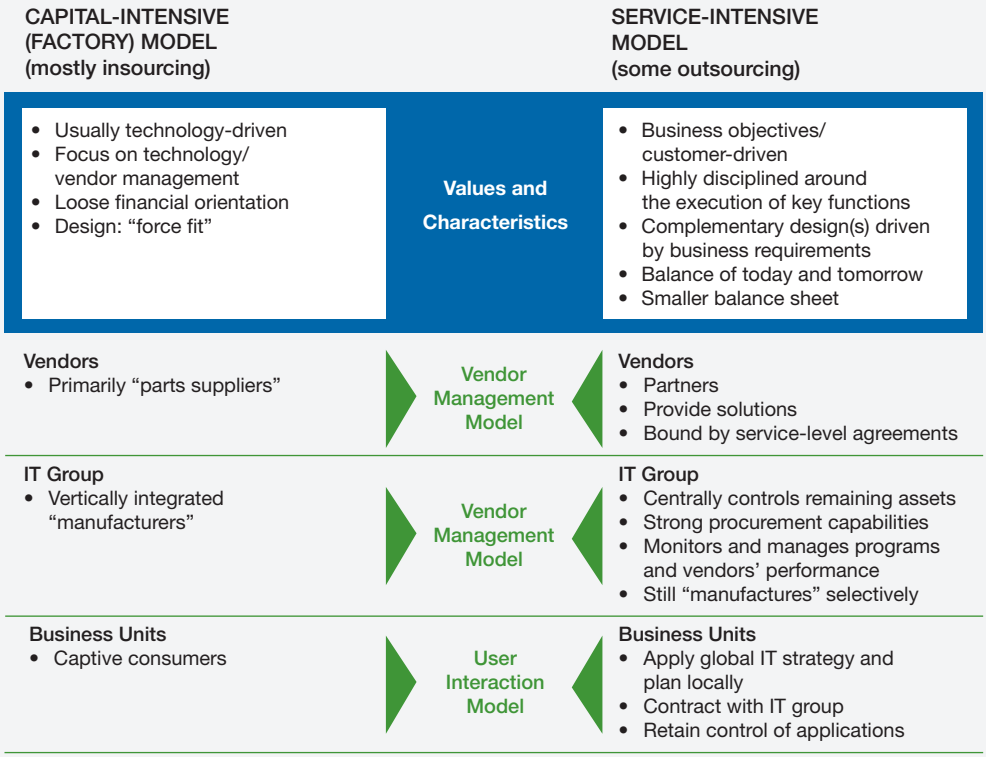
The factory model. In this approach, the organization treats its clients as captive customers who can negotiate only unit cost. It has a balance sheet-heavy approach to resource management: the IT group owns all the resources, including fixed assets, and uses a procurement-driven relationship with its vendors. Although essentially obsolete nowadays, we still find a number of organizations that explicitly or de facto use this conceptual model.

The service organization model. In this approach, which contrasts sharply with the factory model, the IT organization tries to emulate what a service organization does, for example by managing its client relationships through contract-like arrangements and dealing with its vendors in a collaborative mode. In this scenario, a number of resources and assets can be outsourced. A fundamental attribute of this type of organization is that the IT

[10] As we will see in section 6, there is direct, one-to-one mapping between the conceptual, logical and physical concepts and the set of concepts for principle, standard and implementation considerations.

[11] Find more on the definition and usefulness of the IT value chain in section 4.

FIGURE 5 Two Very Different Conceptual Views of the IT Organization



The various combinations of options leads to conceptually very different types of organizations.

organization uses a fairly sophisticated "product management" process to define what it does and present to the user community a menu of solutions in a formal way. Overall, this IT organization adheres to the "service-oriented" model.[12]

The point here is that agreeing to a conceptual vision of the IT organization is a prerequisite to the next step of the organizational design process:

Logical view. Defining the logical view of the organization is fundamentally equivalent to defining the IT value chain, its components and how they interact. This is discussed in more detail in sections 4 and 5, regarding IT value chain and use cases, respectively.

Physical view. Fundamentally, organizational diagrams are the physical views of the organization. Sure, the devil is in the details, but starting at this level of specificity is not the right way to analyze or redesign an organization.

Note that a number of different physical views can be derived from the same logical view. For example, an IT organization's logical view might make a distinction between two separate functions, such as strategic vendor management and procurement, which are in fact very different, even though obviously related.[13] In practice and because of many different factors, these functions might be supported by the same group of individuals.

[12] The concept of service orientation is further discussed in section 9.

[13] Procurement should not be confused with vendor management. Both are needed and overlap, but they have different roles and ideally should be performed by different entities. Procurement should focus on tactical transactions and process, unit cost, and generic terms and conditions. Vendor management has to focus on gathering market intelligence on the supplier market and defining the strategic value of the relationship with key vendors to make their value chain complement and mesh with IT's value chain.

FIGURE 6 From the Conceptual to Logical to Physical Views of the IT Organization

	ABSTRACTION LAYER		
	Conceptual	**Logical**	**Physical**
Focus:	Values, vision, mission	Value chain, processes	Function, reporting, job description
Driven by:	Principles	Standards	Implementation considerations
Managed/ facilitated by:	Management	Collaboratively defined	Facilitated by HR (job definitions)

Figure 6 summarizes the principle attributes of the CLP views of the IT organization.

Summary of Key Ideas

- In developing an organization or a technical design, the temptation to start by drawing boxes and reporting arrows is enormous. This physical view of the organization is, of course, of great importance for all involved, but it cannot and should not be developed without prior, viable conceptual and logical views of the organization.
- The conceptual view of the organization is one that has to be proposed by executive management and reflect its vision and values for the organization. Even though this view might look abstract and almost simplistic, experience shows that arriving at a clear consensus for it is no small task and the exercise of getting to that common vision is fundamentally necessary.
- As we will see in section 6, distinguishing between defining the conceptual view of the IT organization is tantamount to defining key principles that will preside over the design and operation of the organization.
- The logical view of the organization is really where key processes are defined and sections 4 and 5 of this point of view provide a set of tools for that purpose.
- Translating the logical view into the physical view is then almost a straightforward exercise. This said, going from the definition of the conceptual to the logical and then physical view is seldom a waterfall exercise. In practice, all three CLP views are defined somewhat iteratively. This is why it is all the more important that many compromises be achieved in the process. For example, two functions, however distinct, can end up being performed by the same people just because of availability and practical considerations, such as the fact that people have multiple skills. Having several processes supported by the same group is in itself not an issue.

The value chain is useful as a quick and sharp view of the business under consideration within its environment, including suppliers, clients and even competition.

- Another reason for an iterative, pragmatic approach to organizational design is that employees are understandably driven to demand that they know what the transformational process holds for them, which means that management needs to provide regular updates of how the overall exercise is converging toward a concrete organizational design.[14]
- Of all the constructs used by architects, the notion of layering issues into a stack is a very familiar one. Using the concepts of CLP with technologists is sure to elicit at least intellectual interest and most likely a sense that the leadership promoting change in the organization is attempting to do it in an acceptable and methodical way.

4. USING VALUE CHAIN ANALYSIS

What Is It?

A value chain provides an overview of the business by depicting major processes that collectively generate value for the organization and its clients. A value chain is a static view of the business that changes little over time, and you should be able to depict the business in a single view. The value chain is useful as a quick and sharp view of the business under consideration within its environment, including suppliers, clients and even competition.

The analysis allows deeper study of the organization to determine the added value of each organizational component and, most importantly, helps identify the value creation areas.

Examples in the IT World

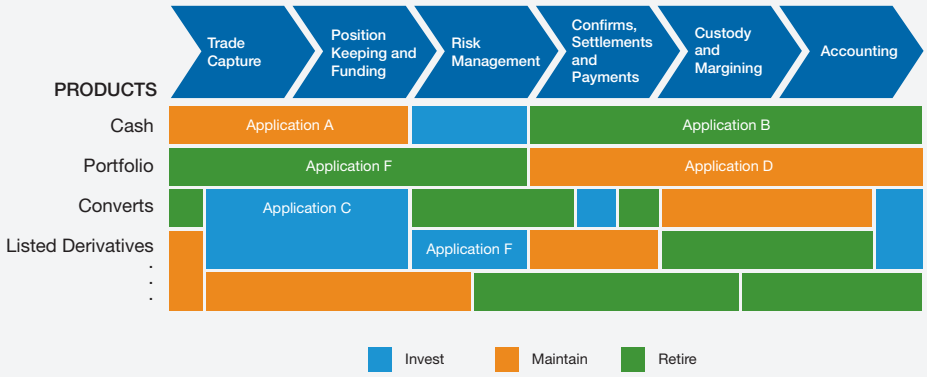
The objectives of developing a value chain from an IT architecture perspective can be defined as:

- Creating a single overview that depicts the major elements of the business.
- Providing an anchor point that can be used as a launch pad for more detailed views from a business, financial and technical perspective.
- Summarizing existing information regarding the "emergent" architectural condition of the organization.
- Defining a common notation and vocabulary for more detailed business and technical architecture work.

[14] See "Deal With Implementation Considerations All Along" and "Early Definition of a Candidate End State is a Must" in the "Architecting in the Real World—Key Performance Factors" section in the BearingPoint point of view, "The Renaissance of Enterprise IT Architecture."

This example for an investment bank shows that the overall enterprise value chain has been broken down into a number of product-dependent sub-value chains for which an inventory of existing applications is depicted prior to a portfolio rationalization exercise.

FIGURE 7 Using the Value Chain Construct in the Context of Application Portfolio Rationalization



Value chains are used in a number of situations, for example, to help develop an inventory of applications and to start a portfolio rationalization process. Figure 7 illustrates an example for an investment bank in which the overall enterprise value chain has been broken down into a number of product-dependent sub-value chains for which an inventory of existing applications is depicted prior to a portfolio rationalization exercise.

Value chain development is also the starting point to identify and develop use cases (see the next section).

Application to IT Organizational Design

Organizations should consider how their IT group(s) uses the value chain construct. This is a very powerful way of looking at what the organization delivers and how it operates. It is of tremendous help to examine the myriad issues associated with all aspects of the organization design and to answer key questions, including:

- What are the key processes?
- How do key processes interact?

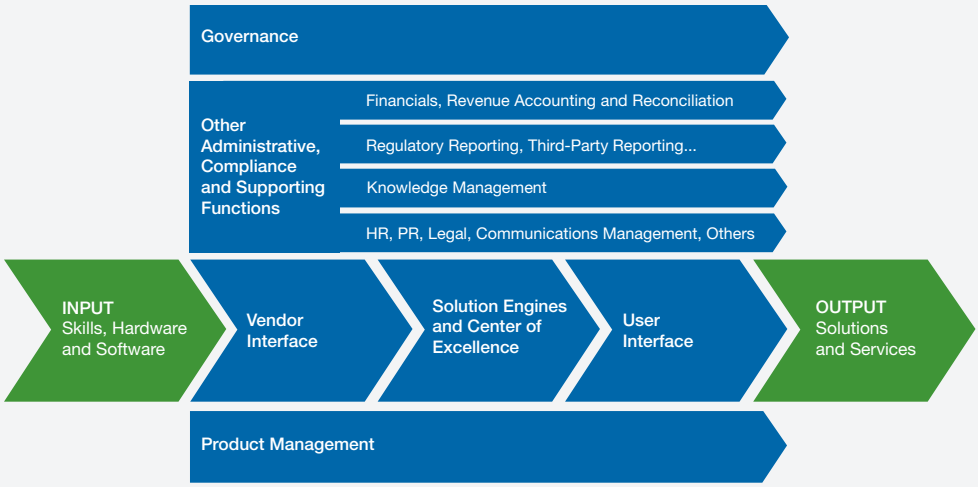
At the highest level, you can represent the IT value chain as shown in Figure 8. Using the concept we defined earlier in the paper, this is a logical view of the organization. This sample diagram indicates that the IT organization, regardless of its actual structure, is designed to "consume" resources and technology provided by suppliers and to deliver solutions to clients.

At the core of this prototypical view of the IT value chain are three key components, further detailed in Figure 9:

- A vendor interface that deals with suppliers. Typically, this includes the support of the processes associated with the strategic vendor management function, but not necessarily the procurement function, which, as we will see, is part of the overall administrative support function.[15]
- One or more solution engines, which are the components of the organization that consume internal and external resources to engineer and deliver value-added

[15]See footnote 13.

FIGURE 8 Prototypical Logical View of the Value Chain Supported by the IT Organization: The IT Value Chain



This sample diagram indicates that the IT organization, regardless of its actual structure, is designed to "consume" resources and technology provided by suppliers and to deliver solutions to clients.

solutions to users. Typically, and without jumping too fast into organizational design considerations—which, as we will see, are implementation considerations—the solution engines are usually organized by technology, which in general includes:

- The communication engine that deals with voice and data communication, sometimes two distinct engines.
- The data center engine.
- The application development engine.
- The distributed computing or desktop support engine.

These "solution engines" can be supplemented by individual or cross-organizational centers of excellence. As an organization evolves, it needs to focus on building areas of expertise in various technology domains staffed by highly competent technologists who would address some of the more cutting-edge, technology-related activities, complex software issues and product/component intellectual property development models.

An important point needs to be noted here. Even if these solution engines focus on specific domains of the technology stack, they deliver what would amount to (in architectural parlance, further defined in section 9, "Embracing Service Orientation") "common services." For example, the infrastructure engine manages the technology resources needed to support the other solution engines. To break this up across individual solution engine silos would result in diluting the expertise these resources would bring collectively as a team.

- One or several client interfaces, which provide the ultimate contact point with the user community. Typically, these include field support, but they are also a form of account management aimed at managing expectations and understanding user requirements (generally aligned by business units).

The core IT value chain in Figure 8 includes two other components, one rather obvious and the other much less so:

- **Governance.** This component of the value chain is self-explanatory because it deals with the definition and monitoring of the execution of the IT vision, in accordance with the corporate or divisional vision. Along with governance, the various administrative support functions handle other functions, such as human resources and finance, regulatory compliance, and security, and role management that help the organization function.

  We cannot underestimate the importance of developing the right governance model—one that places the responsibility for ensuring that the various components of the model work in a coordinated manner.

- **Product management.** This component of the value chain is often forgotten even at this relatively high level of analysis. Here is the underlying idea: in order to deliver value, the IT organization, or any other organization for that matter, needs to act essentially as a regular service organization that delivers a set of products or services, that is, a combination of technology and supporting activities that the user community is willing to buy.[16]

  All of this requires understanding user demand and making sure that your offering is competitive, organized into a cogent catalog and anticipates demand, among many other factors. In fact, our point of view is that the efficient IT group is a service organization that should have organizational attributes of a full-fledged service organization. This combination of activities and this concept have been used ever since marketing has been considered a key function within the enterprise. The name of this function is product management.[17]

[16] This is true even if the user community consists essentially of captive clients (i.e., if the business units cannot procure IT services and solutions directly from other vendors and are only supported by the internal IT organization).

[17] *Product management* is the generic term used by marketing professionals to characterize the set of processes required to organize and modify the scope and nature of what an organization produces and markets. As indicated in section 9, product management for IT organizations requires a proactive, systematic approach that takes into account user demand, the overall environment of the firm in the IT organization and client domains, including balancing the requirements dictated by the firm strategy and internal production constraints. Since an IT organization produces a set of services and solutions, the temptation is great to use the term *services management* instead. Unfortunately, services management in IT refers to the overall set of activities required to deliver services to users. Services management includes activities as diverse as IT asset, fault, change, demand or service-level agreement management. It also includes activities related to the definition of the catalog of services delivered to end users, something also found in product management as defined in this paper. Also, be aware that the term *product* tends to be interpreted by technologists as a package application. The term *product management* is used in this paper despite those ambiguities.

The same logical design can lead to very different physical implementations depending on which additional principles are applied.

Product management is the discipline that ties in and balances the conflicting requirements of:

- Users, advocated by the user interface functions.
- Production, the various in- or outsourced functions involved in delivering the actual services.
- Vendors, as they are understood by the vendor management function.

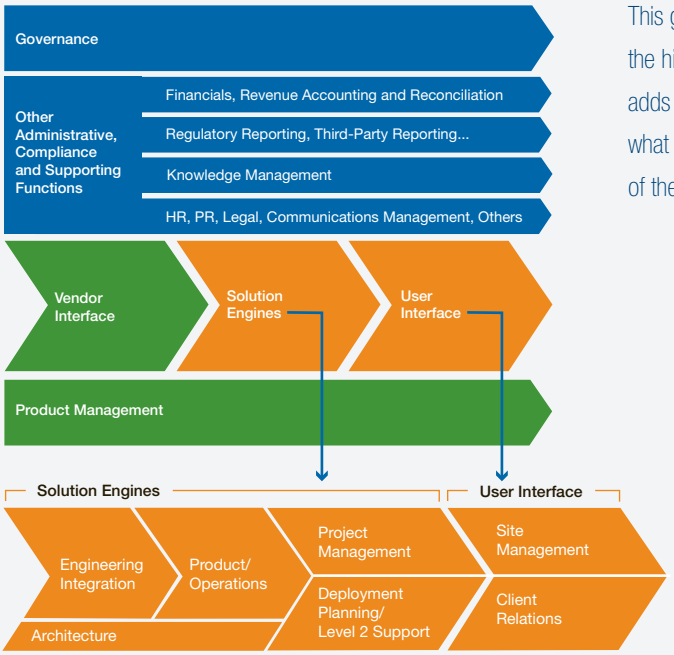
This component of the model deserves a separate discussion, which you can find in section 9, "Embracing Service Orientation." Suffice it to say that we have just accomplished using the value chain construct to start organizing our thoughts in a way that, in combination with the use of the rest of our toolbox, will help resolve a number of issues. In particular, we introduce the concept of product management, which we believe is an essential and often missing part of the IT value chain.

As we observed in the previous section, the same logical design can lead to very different physical implementations depending on which additional principles, as defined in section 6, are applied. Take, for example, the case of an organization that essentially insources all of its IT activity, using a full factory model—that is, the IT organization owns and operates all elements of the value chain. This can be contrasted with an organization that outsources key elements of the solution engines—for example, the operation of its networks. These two organizations use the same value chain model but implement it very differently because different principles are applied.

Detailing the value chain (Figure 9) helps trigger the right discussion as to how each component of the value chain should be managed. This is all the more helpful once we get into a more granular analysis of decomposing each of the broad processes into several sub-processes.

In brief, this granular analysis model takes the high-level view of Figure 8 and adds a level of detail that makes it what can be termed the logical view of the IT value chain because, with this degree of granularity, it is now possible to provide a description of how each component of the value chain interacts. Note that we still have not yet taken this model and attempted to define an actual organizational chart, which will be our physical view of the organization.

FIGURE 9 The Expanded or Logical View of the IT Value Chain



This granular analysis model takes the high-level view of Figure 8 and adds a level of detail that makes it what can be termed the logical view of the IT value chain.

The main additions to the model in Figure 9 are:

- At the user interface, we distinguish between two functions: client relationship and site management, level 1 support. The latter is the classical break/fix field support function. The former is a function that is rarely implemented, or implemented correctly. It represents the activity in which the IT organization interacts with the user community, considered as a client—a function very much akin to account management.
- For each solution engine, we distinguish the engineering/integration, production/operations, project management and deployment planning processes, while defining the architecture processes at that level. The architecture processes are the activities that verify the product management vision is translated into a coherent set of actual deliverables produced by the solution engines from inception at the engineering level to delivery at the operation level.

Summary of Key Ideas

- Value chain representation is a very helpful technique to map out the various processes of the IT organization and to define what each component of the organization does and how it interacts with other components, without introducing the notion of hierarchy or reporting considerations.
- Discussing the value chain of an IT organization, or any organization, makes it possible to develop a reasonably complete inventory of what the organization should do, as well as what kind of interaction it should support internally and externally, in a way that can be made less intimidating for all involved.

- In going through the exercise of developing the target IT value chain, it is important not to merely replicate the current status. In fact, correctly mapping the value chain will require thinking out of the box.
- The proposed reference IT value chain presented in this section is well established to be generic and complete enough to apply to most IT organizations. It can be used as a starting point for discussion. It also scales well, because any subset of the IT organization tends to require the same processes.
- Any subset of an IT organization consumes input from either vendors or other groups and then delivers its output through a user interface. However primitive these processes might be, the notions of product management, governance and others are always present, even for smaller groups within the IT organization.
- The proposed reference value chain highlights the important product management function, which is often ignored by IT organizational planners. Its role and the product management and service orientation concepts are further developed in section 9, "Embracing Service Orientation."
- Value chain analysis is well understood by the business units, and the concept is also widely used in the context of ERP implementation, so technologists are likely to recognize a familiar concept there too.

5. PUTTING THE VALUE CHAIN IN MOTION: LEVERAGING USE CASE ANALYSIS

What Is It?

A use case is a description of a flow of events that provides value to an actor, the user/customer. Use cases capture the dynamics of a business process, as well as the interactions between underlying elements, such as events and services. A use case is documented by providing the name of the participants and describing how they interact in a particular situation.

Examples in the IT World

Take the value chain for a financial firm that focuses on lending to students. As we saw earlier, the value chain depicts the relationship between the major activities and processes. In the value chain in Figure 10, the central chevrons represent the major process of the financial firm, from processing an application to servicing the loan, and the additional threads represent the supporting functions.

A use case is a description of a flow of events that provides value to an actor, the user/customer. Use cases capture the dynamics of a business process, as well as the interactions between underlying elements, such as events and services.

The value chain provides a static view of the business that changes little over time. In contrast, use cases drill down from the value chain and provide information regarding interactions that support the activities. The combination of all use cases provides a dynamic view of the business, for example, the "Apply for Loan" use case helps understand specifically what happens during loan applications.

From a practical standpoint, there are a number of ways to actually document a use case, either in textual form, pictorially or using a combination of both approaches.[18]

A convenient way of showing the actual interaction of actors, either people or system (on top of the diagram), over a period of time (the arrow of time going from top to bottom) is achieved through, for example, what is known as a "swim lane" diagram, accompanied by a textual description of what is going on. Figure 11 is the depiction of the "Apply for Loan" use case.

Application to IT Organizational Design

The relevance of use case as a device to structure any process and test the resilience of a system (defined as a combination of organization, process and technology) to support those key processes has immediate application in the design and implementation of the IT organization (or any organization for that matter).

What we suggest here is that what the organization architect has to do is:

- **Identify the key activities the IT organization has to support.** There are usually two types of use cases: those that directly touch the users and those that are necessary to support the organization itself.[19]
- **Develop the related use case.** For example, Figure 12 presents a list of use cases that were developed to support infrastructure-related activities of a typical IT organization.

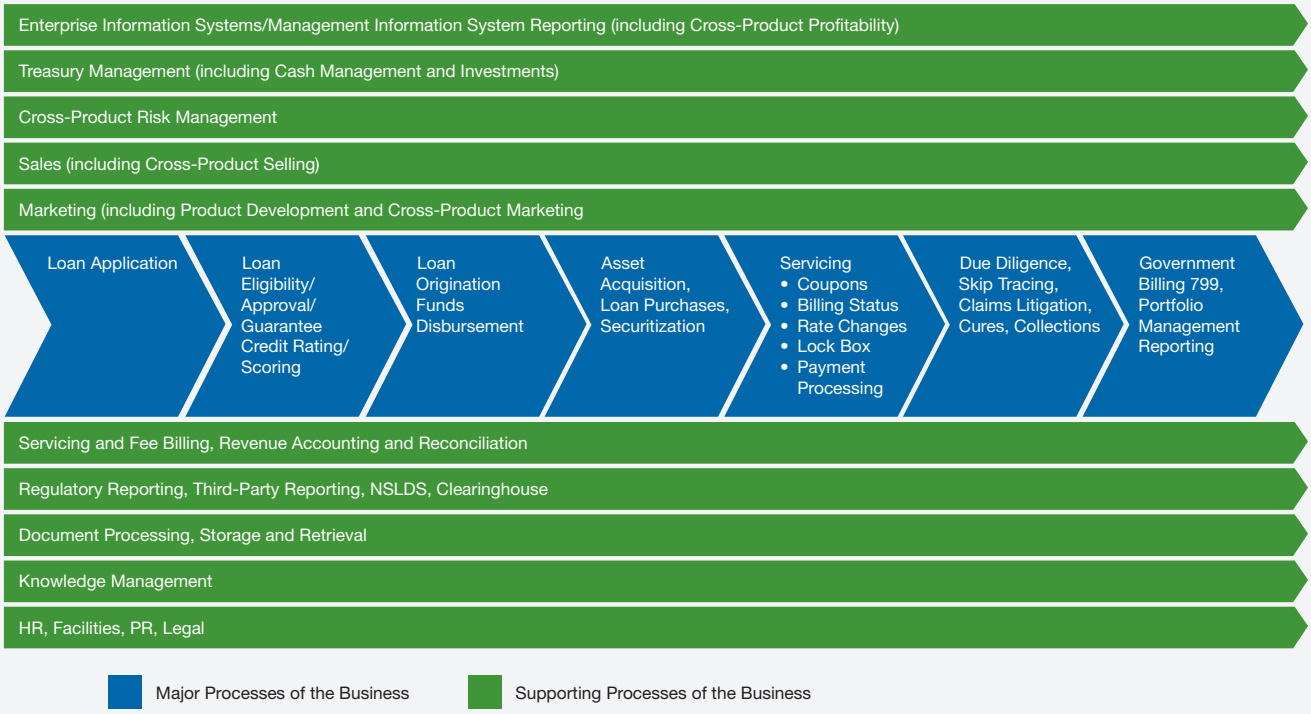
Use cases should be driven by a combination of the business vision/drivers and technology goals of the organization. The joint business vision/drivers and technology goals will help focus on areas of activities that may significantly impact the way the organization needs to be (re)designed.

[18] The set of conventions, tools and constructs together form what amounts to a "visual language," the most popular being the Unified Modeling Language (UML)—the "swim lane" construct used in this paper is a UML concept.

[19] Technologists make a subtle distinction between *scenario* and *use case*, the former being a generic description of the latter. Considering that the term *use case* is somewhat nonintuitive, the term *scenario* might be easier to use with a non-technologist audience.

FIGURE 10 Value Chain for a Financial Institution Focusing on Student Lending (Lend Co.) and Examples of Use Cases

EXAMPLE OF A VALUE CHAIN



EXAMPLE OF A USE CASE (PICTORIAL FORM)

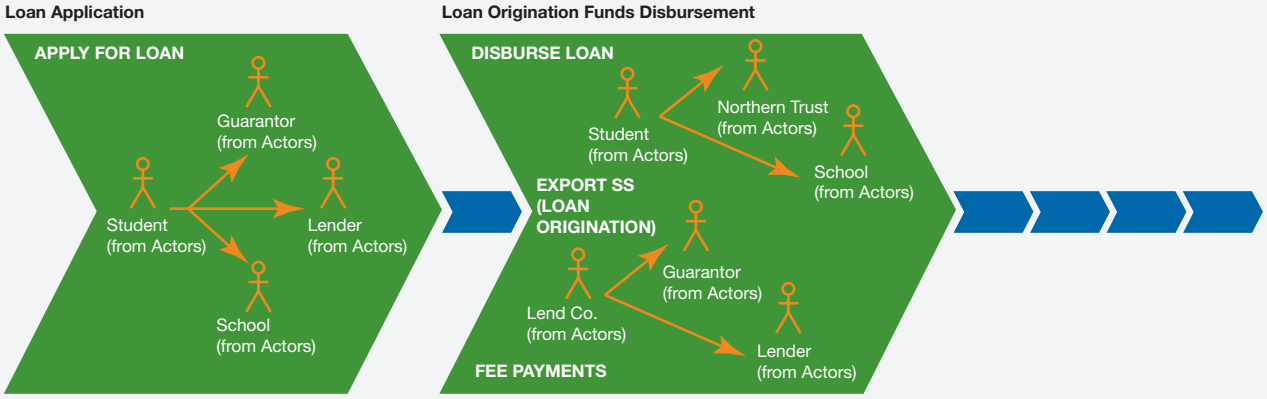
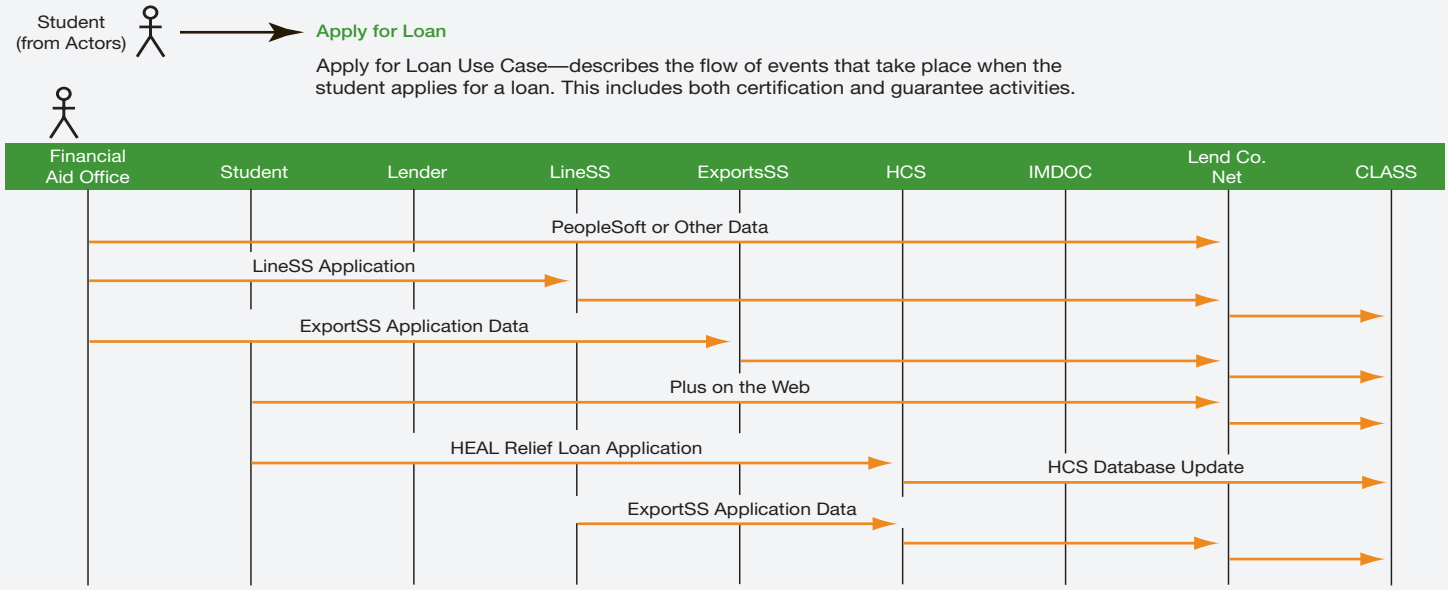


FIGURE 11 Example of Use Case Documentation (Using UML)

EXAMPLE OF A "SWIM LANE" DIAGRAM



Application typically completed and submitted by the student, school or guarantor. This process can be accomplished through several channels, such as Internet, fax or letter. The following actions can occur in any order and must be completed for loan approval:

- Validate certification information sent in by the school. This is done for student loans only. The certification information consists of financials, school information, loan dates, etc. This information is matched and validated against the student application. The certification information from the school can be received via the Internet, Common-Line or letter/fax, phone or manual application. The lender may request the certification from the school or the school may initiate the process.
- Match loan program entered by student to loan program offered by guarantor/lender combination (student loans).
- Obtain loan guarantee from guarantor, if applicable (electronic or manual). For certain loans, blanket or default guarantees apply that make it unnecessary to get explicit approval from the guarantor. However, the guarantor is informed of the loan (guarantees are not obtained for private loans).
- Obtain credit rating from credit bureau (e.g., non-guaranteed loans). The interaction with the credit bureau can be electronic or manual. Input to obtain credit bureau report is done manually. Output is generated electronically online or in the form of a paper report.
- Perform risk analysis and assign rates.
- Obtain underwriting (non-guaranteed loans).
- Perform exception processing, such as credit rejection, application changes, missing borrower information. All exception processing activities have to be communicated to the student, lender and guarantor where appropriate.
- Obtain final approval and inform appropriate parties or notify borrower of denied loan.

Employ use cases to diagram how key activities should be supported effectively by identifying the actors (groups, systems) involved and diagramming out the interactions. For example, Figure 13 shows how the "Open a New Site" use case is implemented using the IT value chain from Figure 12.

Finally, in addition to helping define various organizational and user activities, use cases are helpful in the definition of mutual commitments between actors in the form of service-level agreements—for example, the procurement of desktops should take two weeks, servers would take three weeks, installation takes one week, and so on.

Summary of Key Ideas

- Use case analysis is a concept very well known by technologists. The use of use cases to analyze how their own organizations function is usually very well accepted by that community.
- Just like in the design of architecture in general, selecting the right use case and the number to review is somewhat of an art, but typically the analysis should be kept to a few dozen critical examples.

FIGURE 12 Sample IT Use Cases Set (for Activities Related to IT Infrastructure, Including Networks, User Computing and Data Centers)

Client Relationship	Site Management	Computing	Networking	Governance	Global Deployment
Unplanned business initiative Planned business initiative Unplanned outage Planned outage Budget process Technology deployment Unsatisfied user Over-budget project	Open a new site Complete a user iMAC Convert an existing site New application rollout Half of a LAN segment is down Implement infrastructure project across multiple sites Determine product and functional ownership of new or existing service delivery (e.g., Internet access) Roll out new service Add an employee Distribute or update new software application Train a user on a new software function Build a new PC Respond to a trouble ticket Technology refresh/ new hardware rollout Software distribution	Compliance archiving Application implementation Unexpected volume (e.g., capacity planning and scaleability) Significant fault scenario (e.g., data center disaster, server outage, network outage, haywire application) Periodic reporting Software maintenance	New campus New autonomous geographic region New service development (e.g., Voice over IP) Brand new technology New global application development	Plan, design, development, implementation, maintenance and sunset of utility application used for corporate purposes Development and implementation of language-specific e-commerce applications Implementation of a new desktop standard Management of new company (acquired by database) using nonstandard operating system as an exception Business continuity planning/ recovery readiness audit Development, deployment and decommissioning of distributed system	Routine major deployment (e.g., deploy a new desktop antivirus software package) Routine minor deployment (e.g., deploy a LAN infrastructure upgrade from Win2K to XP in Toronto) Nonroutine major deployment (e.g., deploy Oracle financials) Nonroutine minor deployment (e.g., deploy business unit-specific workstation)

6. DISTINGUISHING BETWEEN PRINCIPLES, STANDARDS AND IMPLEMENTATION CONSIDERATIONS

What Is It?

Architects segregate issues by making a distinction between principles, standards and implementation considerations and have the associated decisions made by different people, at different times, using different approaches.

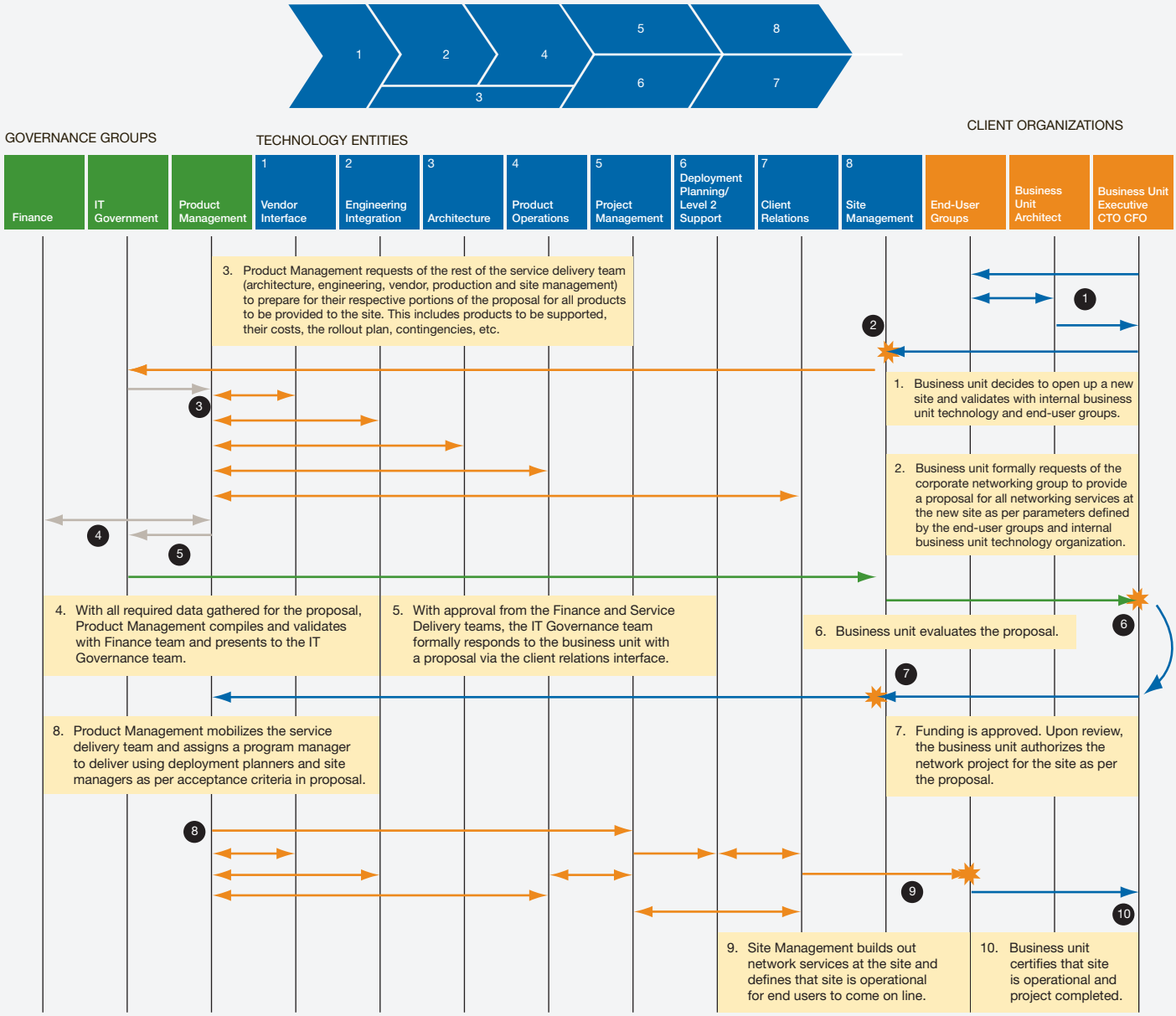
Examples in the IT World

Two examples of this approach in the technology realm are:

- In the system integration realm, deciding to use open source architecture can be construed as a first-level principle, adopting a general SOA approach to architect applications is essentially tantamount to adopting a selected subset of the alphabet soup of SOA-related standards (SOAP, UDDI, WSDL), and implementation considerations would then include deciding whether to use pure open source tools or commercial implementation packages in the field.[20]
- An organization might decide that, as a principle, the enterprise should use a common shared infrastructure to communicate, using TCP/IP as a common set of protocols for a standard. Selecting specific equipment vendors becomes an implementation consideration.

[20]Note that the latter decision (open source versus commercial solution) could itself be decided by another principle that could be adopted by the enterprise (i.e., using only commercial technologies). This illustrates that all components of the principles/standard/implementation consideration stack are interrelated both vertically (several principles might have to be invoked to make a decision) and horizontally (principles cannot be mutually contradictory).

FIGURE 13 "Open a New Site" Use Case



In practice, principles have to be articulated at the onset of many projects, with a scope and degree of specificity/applicability commensurate with and relevant to the problem at hand. For example, a new CIO might start his tenure by proposing and promoting very high-level principles aimed at energizing the organization and making sure that these principles are consistent with the organization's objectives and values.

The distinction between principles, standards and implementation considerations is also necessary to implement the governance model for the IT organization, in essence:

- **Principles**, just like the conceptual view of the organization, should be promoted and enforced by the executive level of the organization and should rarely be modified. Enforcement is usually performed through an ad hoc committee (the IT steering committee) convened by the CEO or CIO, managed by the CIO, and involving business unit leaders as key stakeholders, helped by each business unit technology leader (if such a position exists).

  These principles will have to be correlated explicitly to business benefits for the company to gain from their adoption[21] and should be validated by the IT steering committee. Typically, such a group should make decisions such as "There will be no more than a limited number of database standards throughout the firm" or "We need a corporatewide WAN."

- **Standards** are best left to the community of technology architects throughout the firm, organized into a distributed virtual team (or communities of interests) taking their directions from the principle-setting group. Led by a senior architect reporting to the CIO (or one of the key technology leaders for a business unit most impacted by the standard under consideration), this standards committee should edit and publish reference documents that can then help the implementation community define the specifics of the implementation.

  For example, a standards committee involved in system integration would publish a developer guidebook with the following information: recommended (and deprecated) patterns, criteria for tool selection and use, construction guidelines for applications and wrappers, cross-environment integration techniques, and examples.

[21]See section 10 on quality function development (QFD) as a tool to develop this correlation.

Defining an organizational architecture/model requires the discipline usually associated with the architecture of complex systems.

- **Implementation considerations** should be adapted for various groups involved in the implementation of the standards.

  In the system integration example, implementation considerations include the selection or development of workgroup repositories, construction tools, editors, linkers, compilers, debuggers, emulators, the selection and management of libraries (DLLs, OCXs, classes, templates, etc.), selection and setup of the testing tools for scripting, results capture and analysis, software change management (includes configuration management), and process and project management.

  Note that all of the above could change from project to project, that is relatively often, something that management often resists, but programmers insist on because they believe in "using the right tool for the job."

Application to IT Organizational Design

Defining an organizational architecture/model requires the discipline usually associated with the architecture of complex systems. This requires the definition of the principles, standards and implementation considerations that collectively define an architecture. As indicated in footnote 10, there is one-to-one mapping between the conceptual/logical/physical concepts and the principle/standard/implementation considerations concepts:

- **Principles** govern decisions made and actions taken. Just like IT principles preside over the definition of IT conceptual architectures, organizational principles preside over the definition of the conceptual view of the organization.

  For example, principles related to the use of outsourcing influence the positioning of the IT organization along the factory/service-oriented continuum presented in section 3. The development and acceptance of principles then influences the standards selected.

- A **standard** in IT is a common or formally adopted method for accomplishing something. Various standards may exist, and the choice of one over another is determined by the principles adopted by an organization.

  In terms of the IT organization, standards preside over the development of the logical view of the organization.

  For example, one organizational standard can be to always organize a competitive bidding process in dealing with vendors above a specified threshold of expense.

From a practical standpoint, the notion of standard is closely related to the concept of guideline. Guidelines are a subset of the standards that the organization strives to achieve, although sometimes exceptions need to be made. The organization should capture these and document exceptions wherever they diverge from the guidelines so that they can revisit it constantly as it evolves. This type of situation often appears in the applications realm in which a guideline could be to the effect that all technologies used should adhere to open standards. This is something most organizations strive to achieve but cannot, due to practical considerations.

- **Implementation considerations** need to be defined to make standards actionable.

  Taking the previous example (use of competitive bidding as a standard), defining the minimum number of bidders and the level at which a competitive bidding process is required (i.e., above $50,000) are implementation considerations.

Case Study 2 illustrates the use of principles, standards and implementation considerations concepts used in the design of a new compensation program.

Summary of Key Ideas

- There is nothing earth-shattering needed for IT professionals to consider principles, standards and implementation (PSI). It is, however, not uncommon for management to overlook how important it is to distinguish between them, thus leading to interminable debates on implementation considerations issues while principles have not been firmed up.
- The direct linkage between the PSI hierarchy and the conceptual/logical/physical provides a way to define a governance model.
- Typically, it is executive management's prerogative to develop a conceptual vision of the organization by articulating key principles.
- Key stakeholders should work together with management in defining the logical view of the organization (using the value chain and use case tools, among others), while human resources should support the logistics of the all-too-important implementation of the results of the organizational exercise. This is somewhat simplistic in many ways—for example, human resources should (and usually has) a major role in facilitating the discussions involved.

CASE STUDY 2: PRINCIPLES, STANDARDS AND IMPLEMENTATION CONSIDERATIONS FOR ARCHITECTING A NEW COMPENSATION PROGRAM

Consider the implementation of a new compensation program (CP) aimed at introducing variable compensation within a population of executives who are currently paid using a mostly nonvariable plan. In architecting the new CP, the organization has to adopt three perspectives (or "views"):

1. **Conceptual.** In order to make sure that the plan accomplishes the business objectives, the new compensation plan architects have to make sure that they spell out the key principles that will be used in designing the plan. In this particular case, some of the principles include (note: the following is not an exhaustive list):
   - **Transparency.** In order to promote fairness and meritocracy, the compensation of all executives will be known to all, as well as reviewed periodically by a group of their peers.
   - **Consistency.** The CP will take into account the key level at which activities are performed by the executives (C-level, corporate managers and line managers) and their actual span of control, managed by business or budget, or other appropriate metrics that reflect the magnitude of the responsibility supported.
   - **Partner-like model.** Both the fixed and variable parts of the compensation will be evaluated using an internal currency ("unit") whose value will fluctuate over time as a function of the overall enterprise's results. In addition, individual executives will be awarded an initial set of units and then additional units as a function of current pay level (for the fixed part of the compensation) and ongoing performance (for the variable part).
   - **Promoting ownership.** The variable part of compensation will be available optionally as a combination of cash and stock, with stock being awarded at a discount to promote stock ownership.
2. **Logical.** Implementing the new CP calls for the specific definition of the constructs, that is the standards that will be used, for example, with respect to the four items above. The plan will make specific provisions such as:
   - **Setting up the compensation committees.** In order to implement the bold principles described above, the logical model of the CP will have to spell out the membership rules and operating model of the one or various committees that will actually set up and review how much units are awarded.
   - **Setting up a banding model.** To achieve the bold principles, specific rules need to be agreed on to define the "level" and "magnitude" of the responsibility to be recognized by the CP. For example, by considering three levels (line, management and executive) and four types of magnitude (for example, budget under control of executives in support functions or sales for those in charge of specific delivery), the compensation of all executives will be mapped into a three by four matrix with distinct fixed/variable compensation schemes.
3. **Implementation considerations.** Finally, the CP needs to specify the details of the model, for example, defining the specific boundaries of the bands defined in the compensation matrix. For example, line-level executives in charge of sales of $2 million to $5 million should be compensated between $150,000 and $200,000, and so on for each of the 12 elements of the matrix.

The case study illustrates that a new governance model has to be used for the definition and management of the new CP, recognizing the distinction between:

- Principles, set by executive management and the board.
- Standards, discussed and developed in collaboration with the stakeholders (i.e., the executives concerned).
- Implementation considerations, mostly managed by the human resources department.

7. USING PATTERNS TO DOCUMENT AND LEVERAGE WHAT WORKS AND AVOID WHAT DOES NOT

What Is It?

At a very high level, a pattern is a recipe that has worked before and is shared so that it can be reused in situations similar to the ones where it has been useful. Patterns help users avoid "boiling the ocean" or "reinventing the wheel."[22]

Often, a pattern has propagated throughout the organization in a viral way; it has been adopted by a large population of users who learned about it from other users. To be successful (e.g., used), a pattern needs a tight description of the problem space it covers, along with a clear how-to description and a good set of implementation considerations, to confirm that it can be packaged for reusability.[23]

To be successful (e.g., used), a pattern needs a tight description of the problem space it covers, along with a clear how-to description and a good set of implementation considerations, to confirm that it can be packaged for reusability.

Examples in the IT World

Since time immemorial, every developer worth his or her salt has had his or her own toolset, be it tested code snippets or well-honed techniques he or she uses systematically.

The concept of patterns has been recognized formally and explicitly in the system integration realm at least since the early 1990s, when developers—within their group, then within their enterprise and finally across the whole community—decided to share their wisdom and results and publish catalogs of useful patterns, essentially a collection of well-documented effective practices.[24]

Today, IT strategy and architecture problems are usually solved through analysis and decision making influenced by what architects have either read or directly experienced. Architecture patterns assist architects in this decision-making process by helping to identify the various options for solving a given architecture problem, understand the variations and work with trade-offs. So architectural patterns have emerged, too. These patterns evolve along with the business problems they address and with architects' own experiences.[25]

[22] Using the 80/20 rule of thumb that 80 percent of a situation can be dealt with when 20 percent of the existing solution is itself a pattern.

[23] See the BearingPoint point of view, "The Role of Patterns in Enterprise Architecture Development."

[24] For an example of patterns used in system integration, see the description of the "model/view/controller" paradigm in "Enterprise Credit Management: Digitization Comes of Age" in Issue 2 of the BearingPoint *Financial Services Technology Journal*.

[25] Patterns inherit features from existing defined patterns; this object-oriented approach to patterns makes them easy to develop and is useful in the context of service-oriented architectures and object-oriented development methodologies.

Application to IT Organizational Design

Organizations often talk about documenting leading practices, using reusable documentation or artifacts, or using standardized processes when designing their organizational model.

Here are a few relatively trivial examples: Take the "task force" approach, an organizational design tool, or "multiple reporting," an organizational design construct.

Both "task force" and "multiple reporting" can be considered patterns (and as such capitalized as "Task Force" and "Multiple Reporting"), in that they are solutions that can be used to address certain problems (i.e., Task Force for decision making and Multiple Reporting to enforce multigroup collaboration or share multifaceted resources).

Organizational architects should also leverage the lessons learned in the technology realm. They should mine and document patterns and define their applicability.

The Multiple Reporting or Task Force pattern results can be useful provided that they are used in the right context, a context whose explicit description should be detailed in the pattern definition. For example, by looking at the historical record, an organization might conclude that Multiple Reporting does not work unless it is initiated by the individual concerned, or that it can only work when applied to an individual below (or above) a certain level of autonomy. Another organization (think Dilbert's world) might decide to stay away altogether from using Task Force based on the historical understanding that it doesn't work.

Overall, organizational architects should leverage the concept of pattern, a construct well understood by technologists to define, organize and document leading organizational practices and standards. In addition, IT architects have at times identified anti-patterns, approaches that are known to fail, so that they can be avoided in the future. Task Force might have proven to be such an anti-pattern in many a company.

Summary of Key Ideas

- Recognizing patterns is a way for any organization to become a learning organization, whatever the topic at hand.
- The notion of pattern has a long history in the field of software development but is rarely used elsewhere as such, although it is essentially another way of defining and documenting leading practices.
- Sometimes, recognizing patterns for behavior, concepts or constructs that do not work (so-called anti-patterns) is just as critical.

Overall, organizational architects should leverage the concept of pattern, a construct well understood by technologists to define, organize and document leading organizational practices and standards.

8. USING AGILE DOCUMENTATION AND LIGHTWEIGHT PLANNING: THE IMPORTANCE OF PLATEAU DEFINITION

What Is It?

Key challenges when defining a viable IT architecture include:

- Requirements and the environment changing considerably.
- Lack of a good baseline of what exists (the current state).
- An enormous amount of uncertainty ahead, which might make the definition of a precise end state an exercise in pure speculative futility.

If we try to design an organization using a waterfall approach to problem solving, we find out that a considerable amount of energy can be spent in defining the current state in a way that is rarely used later on (because the description becomes either obsolete, just too unwieldy or both). Much effort is also spent deliberating as to what the end state will look like, only to realize that there are so many unknowns that just about any end state could be a goal. That leaves the organization in analysis paralysis with no clear idea as to how to get started.

The situation is mildly better if the organization uses an iterative approach, but it will similarly be bogged down if it insists on thoroughly documenting the current state and defining the end state with precision. In both cases, defining a real transition plan is practically impossible.

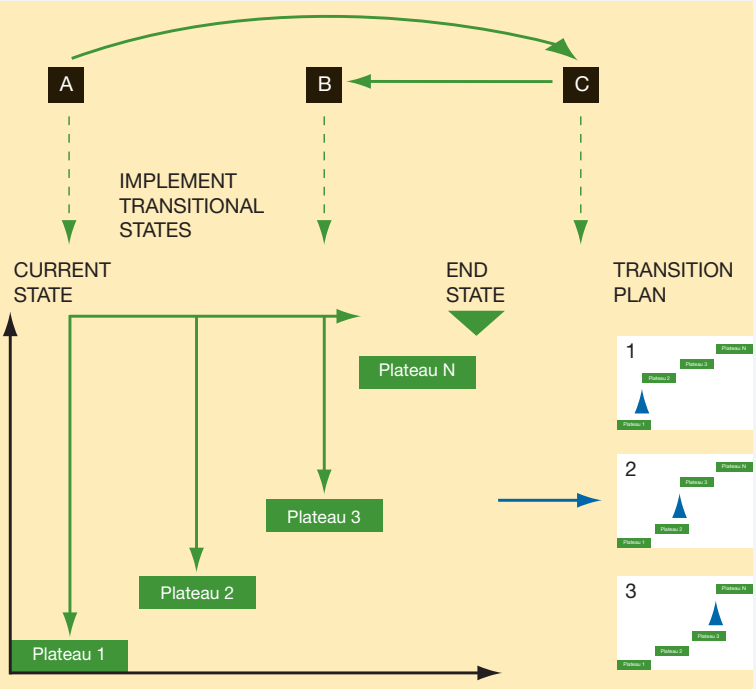
Instead, for an organization that wants to get from A to C, we recommend using a progressive and pragmatic approach that defines realistic "transformation plateaus" (B1, B2, B3, etc.) aimed at implementing the end state step by step, while reducing technology and implementation risks (Figure 14).

Here is how it works:

- **Spend minimal time defining the current state.** Since most of this will be changed and the initial changes will be incremental, everyone involved will start with a pretty good idea of where they stand without having to fill large binders of descriptive information that will remain shelved and unused going forward.
- **Define a high-level view of the end state, but stay at the conceptual level and refrain from going to a higher level of precision.** The information needed to achieve precision is probably nonexistent. In some cases, it is known that the technology necessary to implement that vision is not ready yet, budgets are not available or key talents have not been recruited yet, so the end state can and should remain defined at the conceptual level.

For an organization that wants to get from A to C, we recommend using a progressive and pragmatic approach that defines realistic "transformation plateaus" (B1, B2, B3, etc.) aimed at implementing the end state step by step, while reducing technology and implementation risks.

FIGURE 14 Using Plateaus to Reach an Ever-Changing End State



Implementation plateaus are stepping stones allowing us to reach the desired end state in actionable increments (B).

The challenge lies in developing an actionable plan to reach the end state (C) from the current state (A).

- **Define various implementation plateaus along the way, but focus on executing the very first plateaus.** Only the next and possibly the next plateau after that should be defined in detail (e.g., down to the physical level). We know that things are going to change and do not want to create detailed plans that are likely to change for the plateaus further along in the road map to the end state.

Implementation plateaus are stepping stones that help us reach the desired end state in actionable increments:

- Each plateau should be built with technology that exists at the initiation point. No future promises in that regard should be counted on.
- Each plateau should provide a business benefit that must be clearly articulated with metrics and a scorecard to assess the journey while supporting the business case.
- Each plateau should have a duration of approximately three to six months; there should be enough time to deliver, but the end of the tunnel should also be visible from the outset.
- When reaching a plateau, one should reassess the end state (it might have changed) and/or make sure that the new plateau the organization is embarking on will still be directionally correct.

The overall idea is to develop implementation plateaus as stepping stones allowing us to reach the desired end state in actionable increments, with emphasis on the first plateau.

Application to IT Organizational Design

The overall idea is to develop implementation plateaus as stepping stones allowing us to reach the desired end state in actionable increments, with emphasis on the first plateau.

In essence, what we have described is a variation on the "iterative/low ceremony" theme presented in the first section of this point of view, "Understanding 'Ceremony' and the 'Iterative/Waterfall' Trade-Offs," with additional comments on the futility of spending too much time on the description of the current and end state. Note that, in this context, developing a transition plan consists of documenting the activities required to address the gaps between plateaus at each transition.

Summary of Key Ideas

To avoid the anti-patterns that plague a number of reorganization projects:

- **Do not confuse small random walk and incremental implementation.** Most organizations are reluctant to implement radical changes in the organization and tend to make only incremental changes. The issue here is that, unless these are oriented toward an end state, this is perceived, and indeed is, wasted Brownian motion.
- **Do not focus on the present and the familiar.** Spending too much time defining the current state is rarely useful.
- **Recognize that, at times, the future will no longer be what it used to be.** The end state should indeed be defined, but only conceptually. And, most importantly, this vision of the future should be revised periodically, at least every time a plateau has been reached.
- **Avoid analysis paralysis.** Only the next immediate plateau can and should be analyzed in detail.

9. EMBRACING SERVICE ORIENTATION

What Is It?

A service-oriented "construct" is one that essentially functions through a limited collection of specialized services exchanging well-defined information (or goods).

This item of the technical architecture survey leverages a play on words that attempts to draw a linkage between the SOA concept[26] and another concept in the organizational architecture introduced in section 5 of this paper: the notion of equating what an organization produces with a product or service, thus promoting the definition and use of product management concepts in the definition and provisioning of IT-based solutions.

Here is the logical link: Let's first state the obvious: a service is what an entity (a person, an organization, a piece of software[27]) requests or receives when interacting with another entity. A service-oriented "construct"[28] is one that, when looked at on the conceptual and logical level, shows that entities interact through a collection of well-defined, self-contained, (ideally) self-describing interfaces and does not depend on the context or state of other services.

By that definition, anything but the most monolithic constructs use services. A service-oriented construct is one that essentially functions through a limited collection of specialized services exchanging well-defined information (or goods).

SOA in the IT World

An SOA is an architecture where each application module or data repository connects to one another through well-defined interfacing mechanisms. The communication can involve either simple data passing or two or more services coordinating an activity. Some means of connecting services to each other is needed.

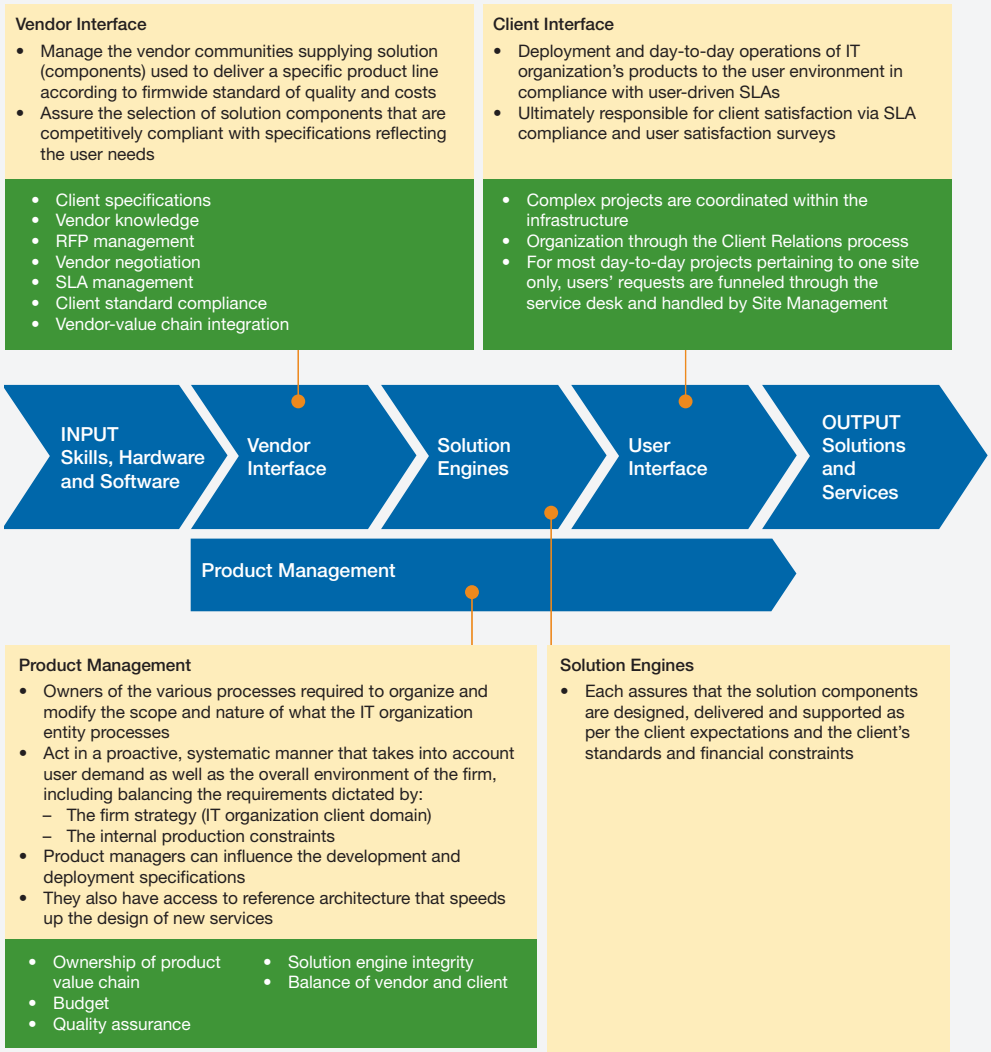
A canonical example of service in the context of application development would be "authentication services." An SOA that implements authentication services provides developers with a common way of making sure that anyone developing a new application that calls for users to log in can insert a simple set of commands in their code that will call a single program (the one supporting the SOA service) anytime the software needs to authenticate the user. This authentication software is developed, maintained and upgraded in a centralized fashion so that software developers do not have to reinvent the wheel every time they need that functionality in their program.

[26] SOA is all the rage nowadays among not only application architects but also less informed but equally enthusiastic business users. Thus our recommendation: If you are promoting a new organizational architecture, make sure you tell management that it's SOA-based and use this section to make it so.

[27] What we referred to an *actor* in section 5 devoted to use case.

[28] A construct is a thing wearing a tuxedo.

FIGURE 15 Some Services Delivered by Each of the Components of the Logical View of the Reference Value Chain Model Presented in Figure 8



If we consider the IT organization as a whole, that means that each functional element of the value chain should deliver specific services to the other elements of the value chain.

There is a substantial amount of literature on the SOA subject,[29] and developing an explication of the concept is itself beyond the scope of this point of view.

Application to IT Organizational Design

The SOA discussion shows something that is intuitively obvious: Entities (be they users or software) tend to interact better when what they get and receive is well defined and obtained through well-defined interfaces. If we consider the IT organization as a whole, that means that each functional element of the value chain should deliver specific services to the other elements of the value chain. Figure 15 shows various components of the value chain proposed in section 4 of this paper and the essential services they provide to one another. In fact, each component of the value chain could be considered a "service."

[29] See "Enabling the Holistic Enterprise With an Integration Framework" in the May 2004 issue of the BearingPoint *Financial Services Technology Journal*.

FIGURE 16 Services Delivered by the Product Management Function to the Other Components of the Organizational Model

Other Elements of the Value Chain	SERVICE DELIVERED AND DESCRIPTION (SAMPLE)
GOVERNANCE *From*	**Guiding principles**—IT executive management (governance layer) provides the product management group with overall principles guiding the definition of the product portfolio and its management. For example: • One of the principles could be using a pricing model that is essentially variable (i.e., business units should be provided with a catalog of services that are priced as a direct function of the volume consumed by the business unit). • Another principle could be that the IT organization should be, in general, an "early adopter" for differentiating technologies, but a "late adopter" in general, or that "buy" is preferred to "build."
To	**Dashboard**—The product management group will provide regular updates to IT management on key metrics, including unit costs. **Competitive assessment**—Most importantly, the product management group should demonstrate how competitive the services offered by the IT group are vis-à-vis solutions and services that could be offered through other means (i.e., outsourcing or by the business unit itself). **Business case**—They should also prepare the business case for investing in new production means by using the "market demand" from the business units (see below) and alternatives in terms of capital investment and return on investment on the overall project.
CLIENT INTERFACE *From*	**Market demand and user feedback**—The client interface is a combination of the service desk, program managers who periodically interact with users and "client executives," who are typically appointed to interface with each of the heads of the business units serviced (Figure 15). Collectively, they should inform the product management function about the value delivered as perceived by the user base and its management (not necessarily the same) and what its future expectations are. *Note that the information provided by the client interface is then used to provide a number of services for the consumption of others (see "User requirements" below).*
To	**Catalog**—The product management group should produce a description of the solutions delivered to the user community.
SOLUTION ENGINE(s) *From*	**Conceptual architecture**—These documents, developed by the solution engines, become the "lingua franca" between product management and the solution engines; this is the high-level description and plan of how the user requirements will be made actionable by the solution engines. **Cost model**—The solution engines present the cost of delivering the conceptual architecture to the product management function.
To	**User requirements**—These are the specific requirements but adjusted to take into account the directives (principles) provided (see build versus buy above). These will be translated into a conceptual architecture by the architecture group.
VENDOR MANAGEMENT *From*	**Product/solution updates**—The vendor management group should consistently inform the product management group on what the vendor can provide. **Competitive assessment**—They also should make competitive assessments of who is best positioned to deliver solutions or solution components to the IT group.
To	**Vendor requirements**—The product management group should describe what kind of solution or solution component will be needed to the vendor management group (the conceptual architecture can be used to that effect).

Of particular importance for this discussion is the role of product management, which influences the other functions at the core of the value chain. Figure 16 shows some of the services that this group should provide to the others. This example also shows that the best way to define what a function does is to identify its interface and define the services delivered to the rest of the functions, an approach that works both for the technology component and organizational design.

Figure 17 shows other services provided at the interface of each of the functions of the model.

Summary of Key Ideas

- Of all the initiatives an IT organization can undertake to truly make a difference in terms of improving the quality and value delivered to the firm, recognizing that the importance of the concept of product management is the one most often overlooked, having the right governance model is key. This is very important to reduce inefficiency, optimize processes and effectively leverage the work done as part of the individual services implementations.

FIGURE 17 Services Delivered by the Components of the Logical View of the Reference Value Chain Model Presented in Figure 9

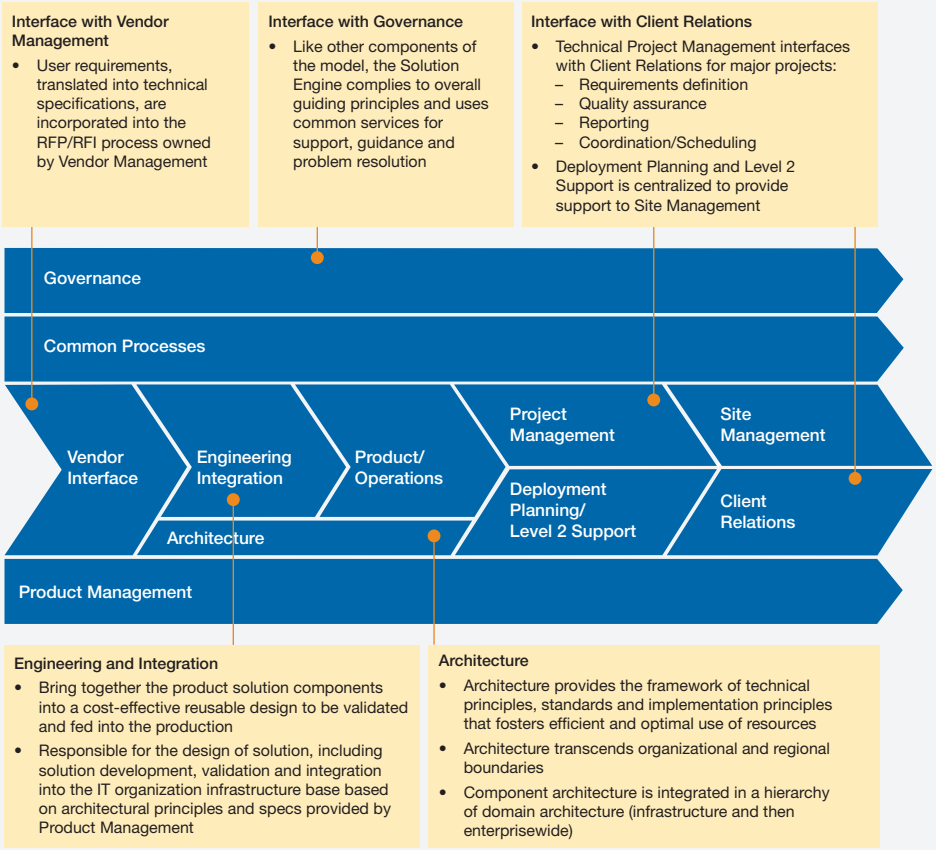


FIGURE 18 Typical IT Solutions/Product Portfolio for a Financial Services IT Institution

PRODUCT LINE	PRODUCT
Access	Third-party (external) access Web services (external clients) Web services (internal clients) Application access services
Hosting	Production hosting per application
Market Data	Market data services
Mobile	Cellular service Wireless messaging services Calling cards Global remote access
Move	Moves, adds and changes (MAC)
Desktop	Desktop/laptop
Solutions	Development tool/instrumentation Computing professional services Telephony professional services Call center voice services
Trading	Turrets Squawk equipment Voice recording services
Media	Media streaming Broadcast services
Voice	Standard voice service Voice messaging systems Auxiliary lines Fax services
Collaboration	E-mail Video conference Workflow, calendaring, scheduling E-fax

- Another aspect of SOA that needs to be emulated is the effective leveraging of existing investments. In the organizational context, we would need to look at how we can leverage the existing organizational structures in the organizational redesign effort.
- Many organizations are embracing a model whereby their product is presented in the form of a catalog (Figure 18). A number of commercial tools available off the shelf help organizations document their product offering this way. It is, however, important to remember that, in order to develop a credible catalog that can also evolve with time, the supporting functions of product management need to exist.
- In essence, IT organizations are service organizations and, therefore, their management should make sure that what they deliver is managed using a product management discipline.

10. LEVERAGING OTHER TOOLSETS AS YOU SEE FIT

What Is It?

There is a great deal of wisdom that can be learned from how organizations deal with issues in fields other than IT or organizational design.

There is a great deal of wisdom that can be learned from how organizations deal with issues in fields other than IT or organizational design. In fact, this tenth recommendation is about leveraging patterns in other knowledge domains. The one that we want to highlight briefly comes from a very different domain: the Six Sigma toolset initially developed in the manufacturing industry. The point here is that, in practice, architecture is no more but no less than a craft that requires tools, and the best craftspeople can spot tools from another discipline and use them to good effect in their specific domain of expertise, whether IT or organizational design.[30]

Examples in the IT World and IT Organizational Design

Even a superficial description of the Six Sigma philosophy is out of the scope of this point of view, but a very brief introduction to how Six Sigma approaches (patterns) and tools apply to technology-oriented and organizational architecture projects is possible.

Generally, the Six Sigma toolset should be considered when dealing with an issue that is neither trivial nor necessarily well defined and/or when the issues at stake are important and cut across organizational boundaries, which is typically true for both IT and IT reorganization projects.

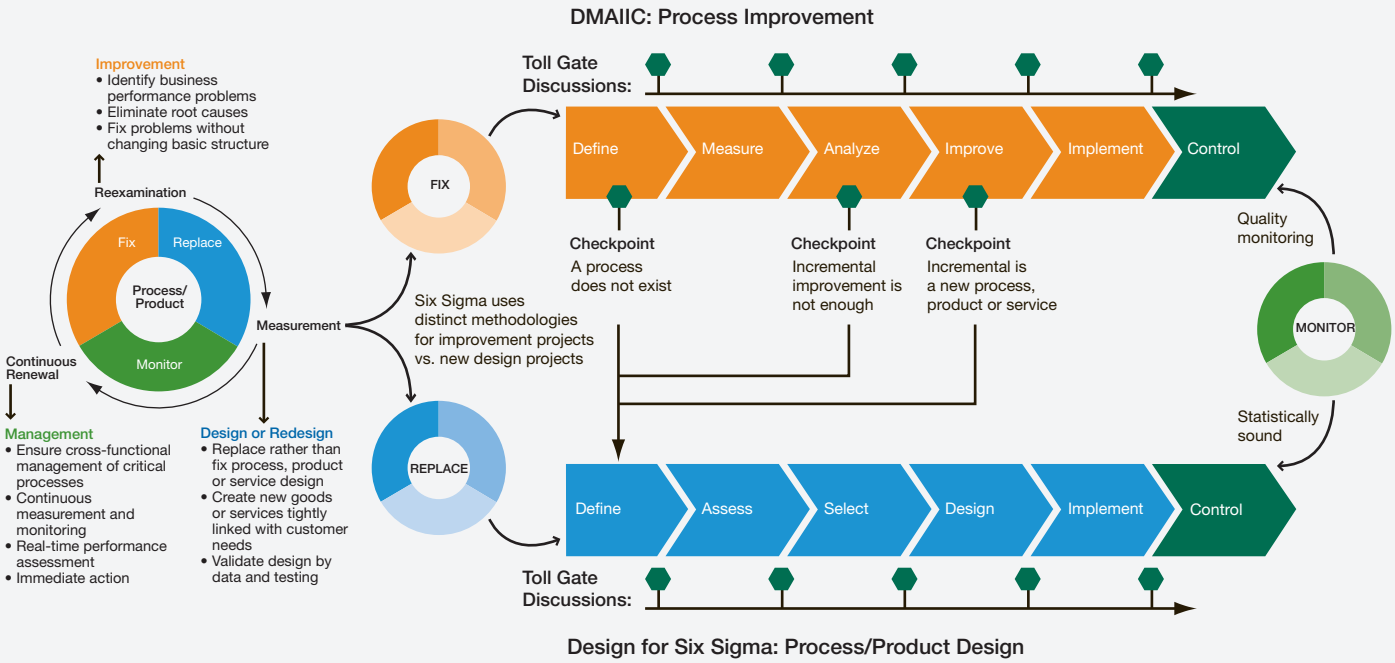
Most people are aware of the use of the Six Sigma concepts and toolset for the purpose of existing process improvement or management, mostly by using statistical analysis and other data-oriented techniques (Figure 19). This is usually accomplished using a combination of key steps—Define, Measure, Analyze, Improve, Implement and Control—referred to collectively as the DMAIIC steps.

Most people do not realize that there is another path through the Six Sigma world, aimed at defining new processes, an approach that can be used if a process does not even exist, if incremental improvement is not enough or if what is needed is an entirely new process, product, service or solution.

[30] For a fascinating example of how to look at problem-solving patterns across disciplines, see *Lattice Work, the New Investing,* Texere, 2000, ISBN 1-58799-00-8, where Robert G. Hadstrom shows how investors can gain insight from the work of others in very different fields.

FIGURE 19 The Various Uses of Six Sigma

The Six Sigma toolbox can be used in three contexts, which require different types of approaches.



In the Six Sigma framework, defining an entirely new set of processes requires a new approach, based on the execution of a new set of different action steps—Define, Assess, Select, Design, Implement and Control—referred to collectively as the Design for Six Sigma (DFSS) approach. As the step names imply, DFSS implies that, in solving an architectural or organizational set of problems, several solutions will be assessed and one will be selected and implemented.

Along with the Six Sigma philosophy comes a wealth of tools that can be used in both contexts, for example:

- **Voice of the Customer analysis.** The "Voice of XXX" approach/pattern helps formalize what key stakeholders "XXX" specifically need and want. For example, in defining a new organizational architecture (or a new compensation/incentive plan, see Case Study 2), a voice of the employee analysis is very important to conduct.
- **Quality function development.** QFD is a general approach that helps both prioritize objectives and trace how solutions can also be prioritized by how well they address the prioritized objectives. By introducing traceability from solutions to objectives, QFD really helps technology and organizational designers focus on what matters.[31]
- **Tollgate analysis.** As is typical for high ceremony/waterfall approaches, DFSS recommends that going from one step to another is subject to the formal presentation of results obtained so far. In practice, the Define phase is probably the most important step in an organizational architecture.
- **Sound Define phase.** DFSS requires that the Define phase include the development of key artifacts, which are really the ones that one would require at the outset of any kind of well-defined project, including:
  - Project mission
  - Work plan
  - Risk management plan
  - Communication plan
  - Return on investment analysis

[31]Stay tuned for an article about QFD in Issue 2 of the BearingPoint *Financial Services Technology Journal* later this year.

By sharing our "universal architecture toolset," we hope to help debunk both the notions that technology architects are the high priests of an inscrutable religion and that organizational design is exclusively a matter of insightful improvisation.

Summary of Key Ideas

- The architecture language is a creole that has absorbed and will continue to absorb contributions from many cultures. This phenomenon has been recognized by technologists, who have, for example, embraced the object-oriented paradigm, initially developed for artificial intelligence and later used in the software development realm. The same approach has been true for the Six Sigma toolset.
- Staying alert to and intellectually curious about new paradigms and tools that can help identify issues and model solutions is something that technology and organizational architects want and need to do to continue to connect the dots.

CONCLUSION

We hope that the description of the 10 approaches we covered in this point of view will benefit both architects and non-architects:

- For non-architects, the point of view can serve as a high-level primer on how to design an architecture, whether a technology architecture or an organizational architecture, and how architects think and work.
- For architects, the point of view can provide a way for them to contribute actively to the debate on how to structure the IT organization.

This point of view is by no means an exhaustive exposé showing the similitude between the two domains. For example, we did not cover the all-important issues related to how to secure funding to implement either. We noted that we know that ROI calculation is not really applicable as an approach to justify investment, but every reorganization effort requires funding. One approach is for both business and technology to jointly fund this effort so that it is seen as an enterprisewide initiative as opposed to an IT initiative.

These 10 concepts are essentially transposable almost "as is" from the IT architecture to the organization architecture realms.

That is not to say that architecting an organization is any less an art than architecting a technology solution is. In both cases, good judgment, experience and intuition have a role to play. But no artist denies the importance of technique in supporting inspiration. By sharing our "universal architecture toolset," we hope to help debunk both the notions that technology architects are the high priests of an inscrutable religion and that organizational design is exclusively a matter of insightful improvisation.

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